

## Day 1: Introduction to PySpark & Databricks Architecture

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### Introduction to PySpark & Databricks Architecture

#### What is PySpark?

PySpark is the **Python API for Apache Spark**, enabling large-scale distributed data processing. It allows developers to work with **RDDs, DataFrames, SQL, Machine Learning, and Streaming** efficiently.

##### ◆ Why PySpark?

- ✓ **In-memory Processing** – Faster than traditional MapReduce
  - ✓ **Distributed Computing** – Parallel processing across clusters
  - ✓ **Lazy Evaluation** – Optimizes execution plans
  - ✓ **Seamless SQL Integration** – SQL queries on large datasets
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#### Apache Spark Architecture

Apache Spark follows a **Master-Slave architecture** with these key components:

- ◆ **Driver Program:** Manages execution, maintains metadata, and converts queries into execution plans.
- ◆ **Cluster Manager:** Allocates resources across nodes (YARN, Mesos, Kubernetes, or Standalone).
- ◆ **Executors:** Run tasks and store intermediate results in memory.
- ◆ **DAG Scheduler:** Optimizes task execution across nodes.

#### Spark Execution Process:

- 1 Create SparkContext
  - 2 Load Data into RDD/DataFrame
  - 3 Apply Transformations & Actions
  - 4 Optimize DAG & Execute Tasks
  - 5 Tasks are distributed & results computed
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#### What is Databricks?

Databricks is a **cloud-based, managed platform** built on Apache Spark that simplifies big data processing.

##### ✓ Key Features of Databricks:

- ◆ Fully managed **Spark clusters**
- ◆ Interactive **notebooks** (PySpark, SQL, Scala, R)
- ◆ **Delta Lake** for improved data storage & reliability
- ◆ **Auto-scaling & auto-termination**
- ◆ Integration with **Azure Data Lake, ADF, Synapse**

## 🔥 PySpark Code Snippets & Explanation

### 🔥 1. Creating a Spark Session in Databricks

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .appName("MyFirstPySparkApp") \
    .getOrCreate()
```

```
print(spark)
```

#### ◆ Explanation:

- SparkSession.builder creates a new Spark session.
  - .appName("MyFirstPySparkApp") assigns a name to the Spark application.
  - .getOrCreate() initializes the session or returns an existing one.
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### 🔥 2. Reading & Writing Data in Databricks

```
# Read CSV from ADLS
```

```
df = spark.read.format("csv").option("header",  
"true").load("abfss://container@datalake.dfs.core.windows.net/sample.csv")
```

```
# Write to Delta Table
```

```
df.write.format("delta").mode("overwrite").saveAsTable("sample_table")
```

#### ◆ Explanation:

- .read.format("csv") specifies the data format.
  - .option("header", "true") ensures column headers are recognized.
  - .load("path") loads data from ADLS.
  - .write.format("delta") saves data in Delta format for better performance and reliability.
  - .mode("overwrite") replaces existing data.
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### 🔥 3. Creating a DataFrame & Performing Operations

```
from pyspark.sql.functions import col
```

```
data = [("Swapnil", 30), ("John", 28), ("Lisa", 25)]
```

```
columns = ["Name", "Age"]
```

```
df = spark.createDataFrame(data, columns)
```

```
df.show()
```

```
# Filter Data
```

```
df.filter(col("Age") > 27).show()
```

#### ◆ Explanation:

- createDataFrame(data, columns) creates a DataFrame from a list of tuples.
- .show() displays the DataFrame.
- col("Age") > 27 filters rows where age is greater than 27.

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### 🔥 Real-Time Scenario & Interview Questions

#### ◆ Real-Time Scenario

🔥 **Scenario:** Your company is migrating its on-premise ETL pipelines to Databricks. How would you migrate workloads?

#### ✅ Solution:

- 1️⃣ Set up **Databricks Workspace & Clusters**
- 2️⃣ Move **data to ADLS/Delta Lake**
- 3️⃣ Re-write ETL scripts using **PySpark**
- 4️⃣ Optimize queries using **Adaptive Query Execution (AQE)**
- 5️⃣ Automate using **Databricks Jobs & Workflows**

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### 🔥 Interview Questions & Answers

#### ◆ Theory-Based Questions

1️⃣ **What is PySpark?**

- ✅ PySpark is the **Python API for Apache Spark**, enabling distributed data processing.

2️⃣ **What are the key components of Apache Spark?**

- ✅ **Driver, Executors, Cluster Manager, DAG Scheduler, RDD/DataFrame APIs.**

**3 What is Databricks?**

✓ A cloud-based managed Spark platform with **auto-scaling, Delta Lake, and interactive notebooks**.

**4 What is a Databricks Cluster?**

✓ A set of **virtual machines (VMs)** that run Spark jobs in Databricks.

**5 What are the different cluster types in Databricks?**

✓ **Standard, High Concurrency, Single Node.**

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**◆ Scenario-Based Questions****6 Your PySpark job is running slow in Databricks. What would you check?**

✓ **Shuffle partitions, DAG optimizations, caching, and cluster scaling.**

**7 You need to integrate Databricks with ADLS. How would you do it?**

✓ **Use `dbutils.fs.mount()` and OAuth-based authentication with Key Vault.**

**8 Your Databricks cluster takes too long to start. What could be the issue?**

✓ **Cold Start Delay, large dependencies, or high user concurrency.**

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**◆ Project-Based Questions****9 How would you build a real-time streaming pipeline in Databricks?**

✓ **Use Structured Streaming, Kafka, and Delta Lake for real-time ETL.**

**10 How do you implement data governance in Databricks?**

✓ **Use Unity Catalog for access control and compliance auditing.**

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**◆ Optimization-Based Questions****11 What is Databricks Autoscaling, and how does it help?**

✓ **Automatically adjusts cluster size to optimize performance & cost.**

**12 What are the best practices for logging in PySpark?**

✓ **Use `spark.log4j` and store logs in ADLS/Synapse for debugging.**

**13 What are the advantages of Delta Lake?**

✓ **ACID transactions, schema evolution, and time travel for rollback.**

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